

# HAMZA MERHABENE

## Geomatics & Surveying Engineer | GIS & Geospatial Engineer

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### PROFESSIONAL SUMMARY

Geomatics and Surveying Engineer with 4+ years of experience across GIS and spatial analysis, topographic surveying, drone photogrammetry, LiDAR processing and remote sensing. Delivers site suitability studies, constraints mapping, cartographic outputs and validated spatial databases for infrastructure, land-development and mineral-tenure projects, and automates repetitive geoprocessing with Python / ArcPy. Currently completing a Master's in Data Science in Lisbon.

### CORE SKILLS

**GIS & Spatial Analysis:** ArcGIS Pro, ArcMap, QGIS, FME, Global Mapper; site suitability, constraints mapping, thematic mapping, topology correction, CRS and datum transformations

**Land & Tenure:** cadastral surveying, land subdivision, tenure and title cross-referencing (BC Crown tenure, LTSA, PLSS / BLM), mining claims, GIS database maintenance

**Surveying:** GNSS RTK, total station (GeoMax), levelling, staking, earthwork and stockpile volume calculation

**CAD:** AutoCAD, AutoCAD Map 3D, Covadis, MicroStation; GIS-to-CAD conversion

**Photogrammetry & LiDAR:** Agisoft Metashape, Pix4D, TerraScan; GCPs, orthomosaics, DSM / DTM, 3D meshes, point-cloud classification

**Remote Sensing:** ERDAS, ENVI; satellite imagery analysis, land-cover classification

**Data & Programming:** Python, ArcPy, SQL, PostgreSQL / PostGIS; HTML, CSS, JavaScript

**WebGIS & Field Data:** GeoServer, Lizmap, Leaflet, QField

**Languages:** Arabic (native), French (fluent), English (fluent)

### PROFESSIONAL EXPERIENCE

#### Freelance GIS Consultant – Land, Tenure & Mining GIS

Oct 2025 – Present

*Self-employed | Remote / International*

- Produce thematic maps, cartographic layouts and technical documentation supporting site suitability, land analysis and tenure assessment across 15+ project areas.
- Map constraints by cross-referencing geological, cadastral and tenure datasets (British Columbia mining datasets, Crown tenure, LTSA title data, PLSS / BLM, mining claims), reducing due-diligence review time by about 40%.
- Convert GIS data to CAD and produce AutoCAD plans, managing coordinate reference systems and datum transformations in ArcGIS Pro.
- Automate repetitive geoprocessing and file-management tasks with custom Python / ArcPy toolboxes, cutting manual processing time by about 60%; run quality control on all geospatial deliverables.

#### Cartographic & Photogrammetric Studies Officer

Oct 2023 – Oct 2025

*WOLO Engineering | Tunis, Tunisia*

- Identified GCPs and georeferenced drone imagery across 30+ survey missions for infrastructure and land-development projects.
- Generated dense point clouds, DSM / DTM, orthomosaics and 3D mesh models in Agisoft Metashape and Pix4D, covering 500+ hectares.
- Classified LiDAR point clouds (ground, vegetation, buildings) and filtered noise in MicroStation / TerraScan.
- Carried out spatial analysis and thematic mapping for territorial planning and site assessment; integrated GIS and CAD data and ran quality control on all client deliverables.

## Surveying & GIS Engineering Intern – Final-Year Engineering Project

Mar 2023 – Sep 2023

GeoTop | Nabeul, Tunisia

- Designed the GIS database of the Kerkennah Island low-voltage electrical network for STEG: data dictionary, UML model (Enterprise Architect), PostgreSQL / PostGIS.
- Collected 1,000+ network assets in the field with QField.
- Processed survey data and corrected topological errors with ArcGIS and ArcPy; loaded the data into the production database with SQL.
- Built a WebGIS for network visualisation and asset management (GeoServer, Leaflet, Lizmap).

## Topographic Surveying Intern

Jun 2021 – Aug 2021

Topography BUT (But Groupe) | Tunis, Tunisia

- Carried out topographic surveys and stockpile volume calculations ahead of quarry operations.
- Performed land subdivision and lot layout for land-development projects.
- Processed survey data, produced topographic maps in AutoCAD / Covadis and computed earthwork volumes for roads and platforms from design lines.

## ACADEMIC PROJECTS – ESAT UNIVERSITY

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- **Flood Risk Mapping** (Oct 2022 – Jan 2023): spatial analysis of flood-prone areas using remote sensing and GIS; risk maps supporting territorial planning.
- **Real-Time Parcel Tracking System** (Feb 2022 – May 2022): design of a delivery-tracking application with real-time geolocation.

## EDUCATION

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**Master's in Data Science** (in progress)

2026 – 2027

Universidade Lusófona | Lisbon, Portugal

**National Engineering Degree in Geomatics and Surveying** (Master's level)

2020 – 2023

ESAT University | Tunis, Tunisia | Grade: Very Good

**Applied Bachelor's Degree in Civil Engineering – Topography and Digital Geography**

2017 – 2020

ISSET Nabeul | Nabeul, Tunisia | Grade: Good

## CERTIFICATION

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Geospatial Programming: Python (ArcPy – ArcGIS) – Udemy